



The Hong Kong Polytechnic University

Project Title: Conducting and monitoring high-precision GPS surveying services in Hong Kong North Metropolis

Tender Reference: FCE/LSGI/4104/Group ■/20■

1. Project Background

In 2021, the Hong Kong government has proposed an ambitious Hong Kong Northern Metropolis plan. This purpose is to redevelop the Yuen Long and North districts. This project focuses on redeveloping cities, which includes constructing housing estates, implementing environmental projects, and developing the necessary infrastructure.

As part of this fundamental creativity, we understand the important requirement of accurate and efficient urban and rural development monitoring. It will use GPS surveying to do this, which will provide essential data and perspectives on the Northern Metropolis's present situation. The planning and construction of important infrastructures and facilities will be greatly helped by this collecting data, which will ensure that they support the overall objective of integrating and sustainable growth in cities.

This tender call aims at finding qualified contractors who may help with the preparation of materials, the beginning of the bidding process, the development of an assessment scoring sheet, and the building of an efficient evaluation framework. Each element of the Northern Metropolis project is important to its effective execution and delivery.



2. Introduction of the Organization (Department of Land Surveying and Geo-Informatics) (LSGI)

(<https://www.polyu.edu.hk/lsgi/about-lsgi/>)

The Department of Land Surveying and Geo-Informatics (LSGI) is an academic unit that specializes in geomatics education and research. It is mainly focused on enhancing the management, modeling, and analysis of spatially linked data with applications for urban and infrastructure development among others.

LSGI specializes in high-assisted monitoring and sensing of different targets that apply high-precision GPS surveying techniques in collecting and analyzing the spatial characteristics of Hong Kong's North Metropolis. The capability to deliver accurate measurements of buildings and other infrastructure linked to ongoing housing construction, and development activities that are supported by internationally trained faculty members and advanced facilities.

Our objective is to connect academic studies with practical applications so that our efforts can effectively support Hong Kong's sustainable development while addressing the requirements of residents.

3. Project Objectives

The objective of this tender is to focus on conducting and monitoring high-precision GPS surveying works within the Northern Metropolis of Hong Kong. This effort's purpose is to ensure that all surveying data to be collected during this effort is at least at the centimeter level, which is extremely important in the context of future urban development plans. With the use of high-precision GPS surveying, sufficient fundamental information will be collected to enable subsequent detailed topographic investigations and 3D modeling to be carried out adequately so that the planning and development of urban infrastructure will be based on reliable and accurate information.



Considering the need for accurate geographic data in the Hong Kong metropolis area, we aim to identify and ensure that certain accuracy and precision requirements are established as goals for the surveying procedures. In addition to enhancing planning and allocation of resources, completing these requirements will support Northern Metropolis's continuing development based on processes in urban areas.

Objectives from this project are going to include the development of 3D models and comprehensive topographic maps, which are important to the key stakeholders in infrastructure, environmental management, and urban planning. This project aims to increase the standards and output of surveying activities for the Hong Kong Northern Metropolis' sustainable development plan, as well as for all administrative and acceptable practice conformance objectives, using current GPS equipment and methodologies.

4. Project Scope

1. **Current Situation Analysis:** Analysis of the different activities of the government concerning the current plans for Hong Kong Northern Metropolis, identifying the main interests, when they are to be implemented, and the expected impact to inform the survey exercise.
2. **Geographical Coverage:** Provide coordinates for the respective geographical areas within the Northern Metropolis where GPS surveying is to be conducted. Provide relevant and specific maps and geographic data that describe these zones accurately.
3. **Key Development Zones:** Select and provide the relevant justification for the key development zones chosen for surveying against the urban design



development proposals, anticipated infrastructure projects, and growth areas contained in government documentation.

4. **Data Collection and Methodology:** List the methods of high-precision GPS data collection including but not limited to equipment used, survey methods, and quality assurance measures to achieve centimeter accuracy.
5. **Collaborating with the Stakeholders:** Explain how you will collaborate with the relevant stakeholders, such as the government, city planners, and the inhabitants, to ensure all the surveying activities are within the context of wider urban development strategies.
6. **Reports and Outputs:** Describe the deliverables which are anticipated outputs of the project, such as comprehensive contour maps as well as three-dimensional (3D) models for effective urban planning, and periodical reports to update the clients during the project.

5. Project Deliverables

1. **Survey Report (following COP v5):** The Code of Practice (COP) version 5 standards shall be followed in this report to ensure that each survey work is fully documented. It will give an in-depth understanding of the surveying procedure and its results by including GNSS control graphs, station summaries, and network adjustment data.

Methodology and Results Analysis Report: This report will detail the methodologies employed during the surveying process, along with a thorough analysis of the results obtained. It will also discuss potential challenges encountered during the surveying activities in the Hong Kong Northern Metropolis, offering insights into future projects and improvements.



2. Detailed Topographic Map: The Northern Metropolis's geographical elements will be accurately represented on a high-resolution topographic map. With its accurate coordinates, this map will be an important tool for infrastructure development and urban planning.
3. 3D Model of the Hong Kong Northern Metropolis: The Northern Metropolis urban environment will be represented through the creation of a comprehensive 3D model. Stakeholders working on urban planning and design projects can benefit from improved understanding and evaluation made possible by the model's specified coordinates.

The project will be steered by Land Surveying and Geo-Informatics (LSGI) Staff with project management by the LSGI Secretariat. The Consultant shall complete the project within **12 months** of the tender award. LSGI reserves the right to edit any items of work subject to mutual agreement of both Parties.

6. Consultant Requirements/Qualification

Competent Project Team: A competent project team with relevant qualifications, knowledge, expertise, and experience in similar geomatics projects related to GNSS surveying is essential for the successful execution of the project with at least 3 years of experience.

Local Researching and Understanding: To ensure that the surveying tasks are suitable for the environment and local development plans, the consultant must show the capacity to conduct local research and have a comprehensive awareness of the Hong Kong Northern Metropolis.

The team shall comprise members with expertise and experience in the following areas:



- GPS Surveying: To ensure effective data collection and analysis, team members must have several years of experience with GPS surveying techniques.
- Aerial Surveying (UAV): Skill in Unmanned Aerial Vehicle (UAV) aerial surveying is required to improve data collection and provide complete geographic insights.
- Topographic Surveying: To produce detailed maps that accurately represent the topography and features of the Northern Metropolis, the team has to have an excellent background in topographic surveying.
- Relevant Software Proficiency: Experience in using relevant software tools, such as Leica Geo Office, Revit, Context Capture, and Cyclone Register 360, is necessary for efficient data processing and modeling.
- GNSS Techniques: Knowledge of GNSS methods, such as Differential GNSS (DGNSS) and Real-Time Kinematic (RTK), is necessary for ensuring high-precision collection of data and improving the accuracy of the results of surveys.

7. Scoring Sheet Development

The scoring sheet has been developed to quantify proposals made by suppliers regarding the “Conducting and monitoring high-precision GPS surveying services in the Hong Kong Northern Metropolis” project. It is also noted that each proposal lodged will be scored by some criteria representing technical capacity, cost estimate, relevant execution, quality assurance, and risk management of the suppliers. This overall score would in turn help highlight the best-suited consultant for the execution of the project considering all factors necessary for the successful outcome of the endeavor.



1. Technical Capability (30 points)

- Accuracy of GPS Technology: assess how the supplier can accurately provide a centimeter-level precision for the GPS surveying scale
- Use of Advanced Tools: estimates the effectiveness of the use of DGPS and other advanced tools to manage signal interference better.
- Experience with Similar Geomatics Projects: narration of the supplier-based experience and expertise in executing geomatics of similar level.
- Adequacy of Equipment and Technology examines whether the supplier possesses the relevant kinds of equipment and technology appropriate for urban surveying.

2. Cost/Price (20 points):

- Competitive Pricing: Evaluate the proposed bid price against the provider's proposed prices on comparable jobs.
- Comparison With Other Bids: how the proposed price can be compared to other bids that will sort out all received bids.
- Comprehensive Budget Coverage: Assess whether the proposed budget will be enough to cater to all relevant aspects including technology and equipment provision, labor provision, and risk management.



3. Experience and Track Record (20 points)

- GPS Surveying in Urban Environments: Focus specifically on the supplier's experience in conducting GPS surveys in urban areas.
- References and Past Project Outcomes: Seek reference letters from previous clients and inquire about the past projects' outcomes for dependability and effectiveness.
- Ability to Meet Deadlines and Budgets: Evaluate the supplier's ability to complete projects within the scheduled times and the allocated budgets.

4. Quality Assurance (15 points)

- Quality Control Processes: Prepare to examine how the supplier manages its business and what quality control processes are employed.
- Data Accuracy and Reliability Procedures: Analyze the procedures that the organization or the individual has established to maintain the accuracy and reliability of the data.
- Certifications and Standards: Examine whether the supplier has any pertinent certifications or follows any standards e.g. the ISO standards to ensure quality.

5. Risk Management (15 points)

- Strategies for Signal Interference: Examine the strategies used by the supplier to combat signal interference and ensure the accuracy of the GPS in unideal urban



settings.

- Emergency planning: Examine the measures that have been proposed to handle unexpected currently anticipated postponements or delays during the project.
- Urban Area Risk Management: Examine the performance of the supplier regarding the risks presented by carrying out the work in urban areas.

8. Instruction to the applicant organization

In the case that there is only one application the company, and interested organizations must submit an Expression of Interest (EOI) focused on a quantitative performance of high-precision GNSS surveying in the Northern Metropolis of Hong Kong. Please follow the guidelines listed below:

- Specific Criteria: Make sure that your organization has the legally required qualifications as mentioned in the EOI including the experience of the organization in GPS and the experience of working in urban development activities which necessitate such skills.
- Content Requirements: The EOI should detail the relevant experience of the EOI applicants including administrative and technical capabilities and how the GPS surveying services will be delivered.
- Clear Presentation: Ensure the information on how the project will be undertaken includes all relevant steps needed for easy and effective assessment of each of the sections' performance.
- Confidentiality: The submission of such information shall constitute an assurance that the information given shall be regarded as confidential and the



information shall be used solely for the evaluation of the submission.

- Contact for Clarifications: In case of any queries or need of further information about this EOI communication please refer to Chan Tai Man as provided in the contacts.

9. Challenges and Solution

1. Boundary Definition Challenge

Challenge: One of the most defining characteristics when the project area has more than one manager area and overlapping areas is the boundaries that take precedence in the management order.

Solution: In this context, we shall work with local governments to learn important things concerning borders of jurisdiction. All these consultations will also aid in resolving overlapping issues and it will be worse for us when we come to establish survey areas that are not only clear but also precise which will take care of compliance and the scope of our work.

2. Signal Interferences

Challenge: The signal of GNSS surveying may be absorbed by surrounding buildings or trees affecting the overall outcome of GNSS surveying data.

Solution: To eliminate signal interference, the project will seek to integrate Differential GNSS (DGNSS) and ground-based augmentation systems into the project plan. This integration of technologies would help increase the level of signal accuracy while limiting the signals that are otherwise invisible. Therefore, data collected for the project will be within prescribed levels of reliability even in urban areas where



obstructions abound.

3. Geographic Location

Challenge: The accuracy of GNSS surveying is negatively affected by the surrounding high terrain and hill shades which have been reported in most cases.

Solution: To enhance the reliability of signals within these regions where geographic challenges are immense, ground-based augmentation systems will be deployed. Also, whenever practical, Aerial surveys will be undertaken to obtain coordinates for points in the shaded relief areas, thus enabling guarantee of data coverage.

10. Ways to submit

Interested consultants are invited to submit their Expression of Interest (EOI) in undertaking this consultancy.

The EOI should include the following information:

1. **Company Profile, Background, and Expertise:** Provide a comprehensive overview of your company, including its history, areas of expertise, and relevant qualifications in the field of high-precision GPS surveying.
2. **Background of the Project Team Members:** Include detailed profiles of the project team members, highlighting their qualifications, relevant experience, and roles within the project.
3. **Previous Relevant Experience/Projects:** Summarize previous projects related to high-precision GPS surveying, demonstrating your capability and experience in delivering similar consultancy services.



The Expression of Interest (EOI) should not exceed 6 A4 pages. Sealed tenders, in duplicate, must display the tender reference and subject on the outside of the envelope (without any identifying marks related to the tenderer). These should be addressed to the Chairman of the Central Tender Board and placed in the LSGI Tender Box, located in Room ZS621, 6th Floor, South Wing, Block Z, Phase 8, 181 Chatham Road South, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong, by **17:00 (05:00 p.m.)** on 19 Nov 2024. Tenders must be submitted to the specified tender box as outlined in this notice before the closing time.

Late submissions and those not placed in the specified tender box will not be accepted. If the event that a Typhoon signal No.8 or above or a black rainstorm, the warning is hoisted on the closing date announced by the Government is/are in force at any time between 9:00 am and 12:00 noon on the tender closing date, the EOI submission closing date will be postponed to 6:00 pm on the following working day.

This invitation for EOI doesn't mean to offer nor does this document or any document made available under this invitation constitute any contract or agreement of any kind

whatsoever with LSGI. LSGI also reserves the right to cancel or terminate the process of EOI at any time and shall not be liable to any party for any loss or damage, cost, or expenses resulting from such action.

11. Evaluation Criteria

Relevant experience in topography surveys: Having valuable experience where detailed topographic surveys will be carried out on similar projects and provide instances of the typical nature of such work to highlight competencies within the same area.

Past performance: Submit the completed projects dealing with topographic surveying and the creation of digital elevation models, including client contacts and results of projects if any at all.



Management and technical capabilities: Possess good management and technical capabilities in topographic surveying, spatial data processing, and any GIS software packages. Skilled use of building information modeling and data amalgamation should be put to maximum use to achieve high-quality deliverables.

Basic knowledge in using equipment: Demonstrate a basic understanding of operating different surveying equipment such as total stations, GNSS systems, and LiDAR which are essential for data collection purposes.

Usability of the technical resources: Provide evidence of the existence as well as use of essential equipment and technology to execute and complete the project promptly and appropriately in all instances meeting the industry benchmark.

12. Enquires

For any inquiries concerning this EOI invitation, please contact:

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Email: lsdept@polyu.edu.hk

Office Hours: 08:45 – 17:35

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